



Patient Profile

Name: Mr. PL94 Age: 35 Gender: Male

Summary of Results

The comprehensive laboratory panel reveals several deviations from reference ranges, notably elevated total and direct bilirubin, a mildly raised AST, and a low alkaline phosphatase level. The predictive model flags the overall profile as **Abnormal**, indicating that the pattern of results warrants clinical evaluation.

Detailed Test Explanations

Bilirubin Total (14 mg/dL, High) – Bilirubin is a breakdown product of hemoglobin. Elevated total bilirubin commonly signifies impaired hepatic excretion, hemolysis, or obstructive pathology. In this patient, the value far exceeds the normal upper limit of 1.2 mg/dL.

Bilirubin Direct (1.2 mg/dL, High) – Direct (conjugated) bilirubin reflects hepatic processing and excretion into bile. A level above 0.2 mg/dL suggests cholestasis or biliary obstruction.

Serum Bilirubin (Indirect) (0.20 mg/dL, Within range) – Indirect bilirubin, the unconjugated form, remains within normal limits, indicating that the rise in total bilirubin is primarily due to the direct fraction.

SGOT / AST (36 U/L, High) – AST is an enzyme present in liver, heart, and muscle cells. Mild elevation above 33 U/L may indicate hepatocellular injury or inflammation.

SGPT / ALT (35 U/L, Within range) – ALT is more liver-specific; a level within 0-41 U/L suggests the injury is not severe or may be predominantly extra-hepatic.

AST / ALT Ratio (1.03) – A ratio near 1 indicates a balanced pattern, not strongly suggestive of alcoholic liver disease (which typically shows a higher ratio).

Alkaline Phosphatase (ALP) (25 U/L, Low) – ALP is released from bone and bile ducts. Low values may reflect malnutrition, hypothyroidism, or bone disease but can also occur in isolated hepatic cholestasis.

Total Protein (6.9 g/dL, Normal) – Reflects overall protein synthesis and nutritional status; within reference range.

Albumin (4.6 g/dL, Normal) – A marker of hepatic synthetic function; normal.

Globulin (2.3 g/dL, Normal) – Represents immune proteins; within normal limits.

Albumin/Globulin (A/G) Ratio (2.0, Normal) – A balanced ratio indicates no obvious protein loss or synthesis abnormality.

Overall Interpretation

The laboratory findings point to a predominant cholestatic pattern, evidenced by markedly elevated direct bilirubin and a low alkaline phosphatase. Mild AST elevation, with a normal ALT, supports the possibility of a hepatocellular component but is not pronounced. The disproportionate rise in conjugated bilirubin, coupled with the low ALP, suggests an obstruction or dysfunction at the level of the hepatic ducts or within the liver itself, rather than a bone-related ALP

deficiency. Normal albumin, globulin, and protein levels indicate that synthetic hepatic function remains intact at present. The predictive model's abnormal flag aligns with the biochemical evidence of hepatic derangement.

Recommendations

- **Clinical Correlation:** Immediate consultation with a hepatologist or primary care provider to assess for signs of jaundice, pruritus, or abdominal pain.
- **Imaging:** Abdominal ultrasound or MRCP to evaluate biliary tree anatomy, gallstones, or biliary strictures.
- **Additional Testing:** Viral hepatitis serology (HBV, HCV), autoimmune markers (ANA, AMA), and iron studies to rule out hemolytic or metabolic causes.
- **Medication Review:** Identify any hepatotoxic drugs or supplements.
- **Lifestyle Modifications:** Abstain from alcohol, maintain a balanced diet, and ensure adequate hydration.
- **Follow-up Labs:** Repeat liver panel within 4–6 weeks to monitor trend; consider more detailed enzyme panels if abnormalities persist.

Disclaimer

This report is for informational purposes only and does not substitute for professional medical evaluation. Please consult a qualified healthcare provider for diagnosis and treatment.